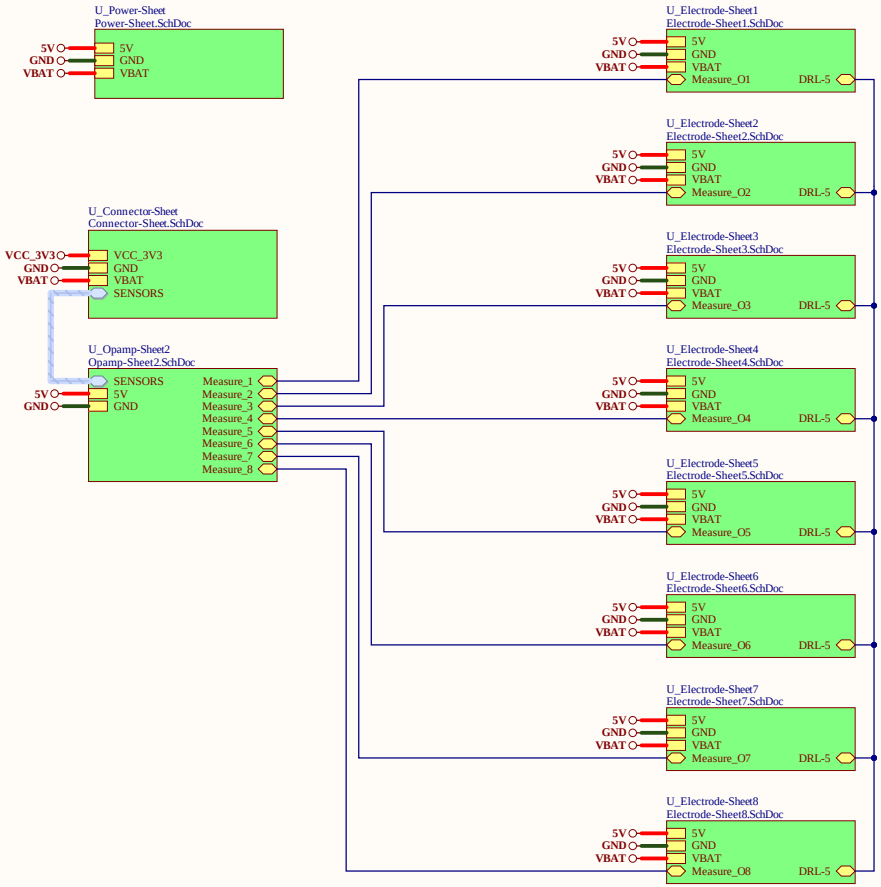
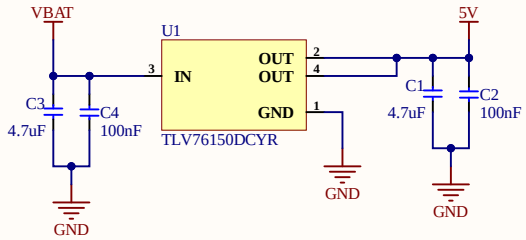
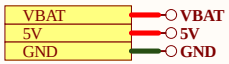



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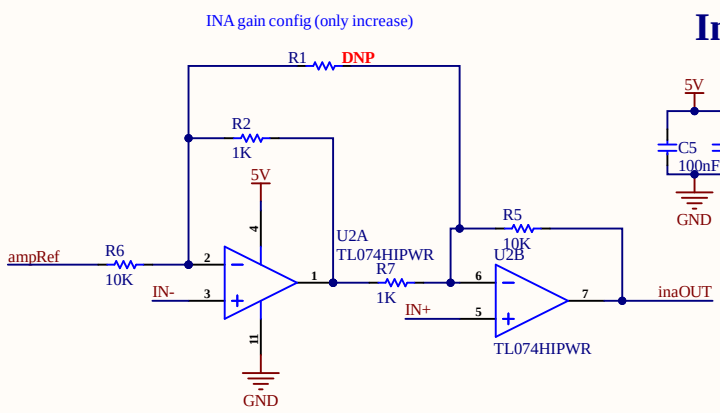
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| Title: Top-Sheet.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/10/2025 | Page Num: 1 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

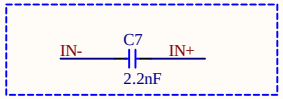



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|---|---|
| Title: Power-Sheet.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/7/2025 | Page Num: 2 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

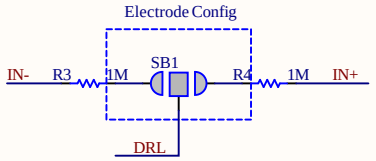
# Instrumentation Amp (INA)



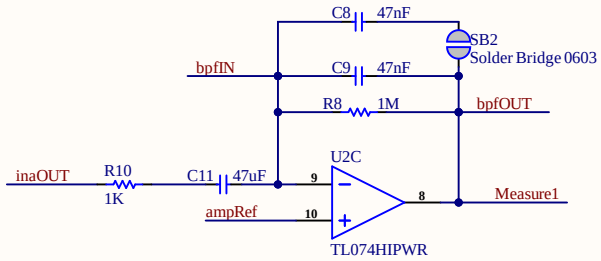

Removes high frequency noise.

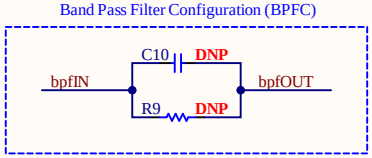
# Electrode Refrence




Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

# 1000x Gain (Bandpass)

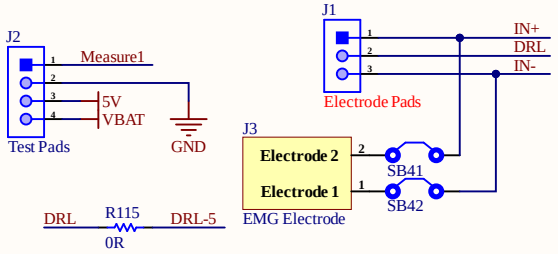


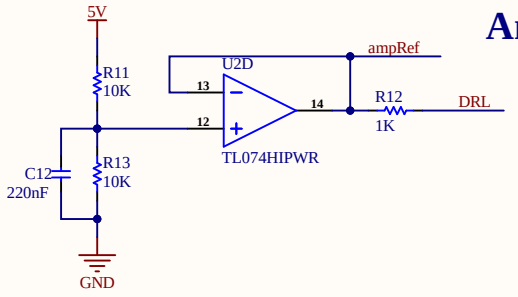

Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

# Header Pins / Connectors

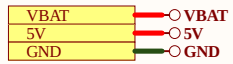



# Amp Ref + Driven Right Leg (DRL)





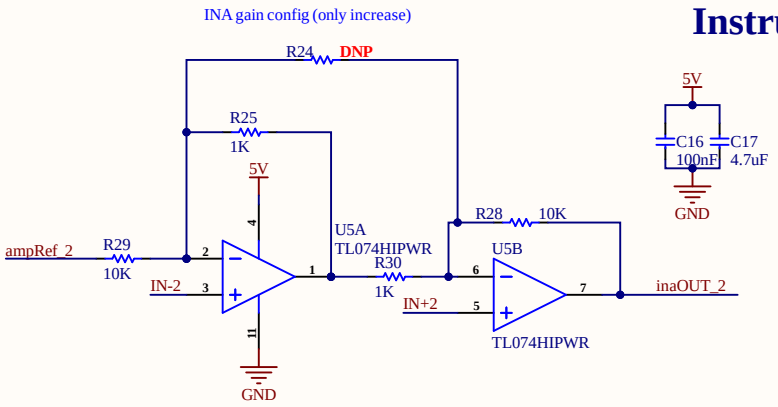


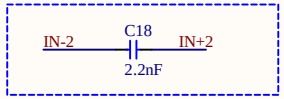



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| | |
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| Title: Electrode-Sheet1.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/10/2025 | Page Num: 3 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

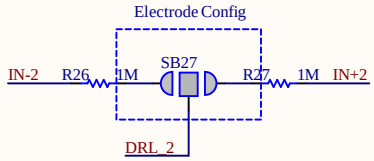
# Instrumentation Amp (INA)



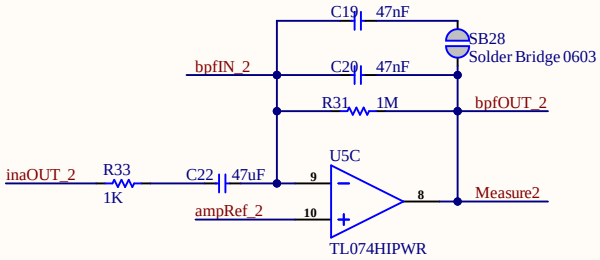

Removes high frequency noise.

# Electrode Refrence

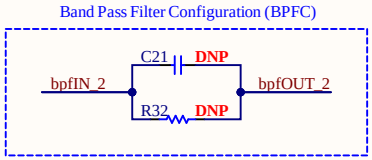



Bridge all three pads of the solder jumper to configure for
2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

# 1000x Gain (Bandpass)

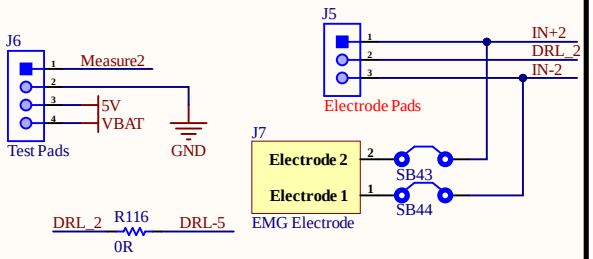



Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

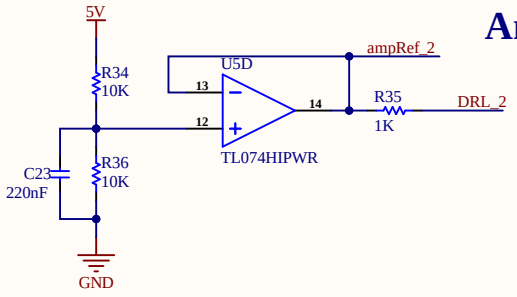



C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

# Header Pins / Connectors

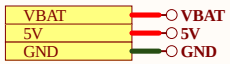



# Amp Ref + Driven Right Leg (DRL)





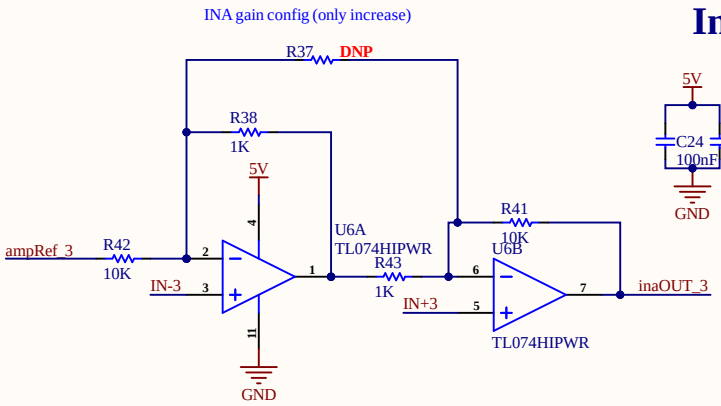


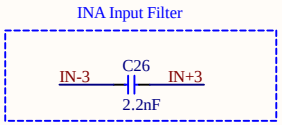



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|---|---|
| Title: Electrode-Sheet2.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/10/2025 | Page Num: 4 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

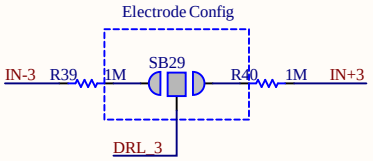
# Instrumentation Amp (INA)



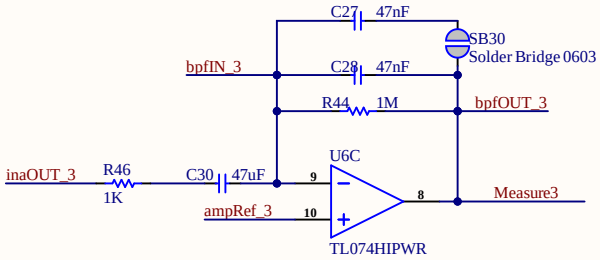

Removes high frequency noise.

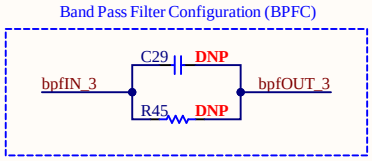
# Electrode Refrence




Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

# 1000x Gain (Bandpass)

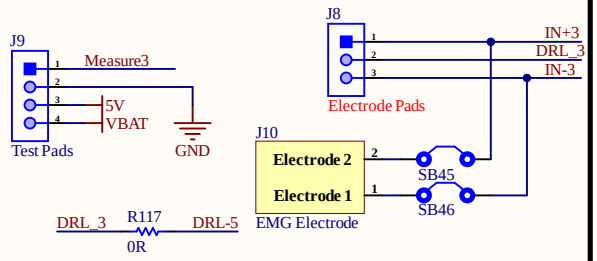


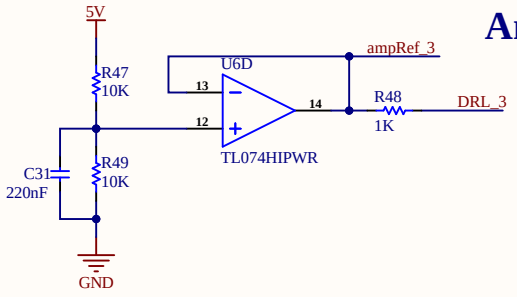

Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

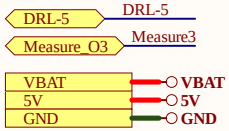
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

# Header Pins / Connectors




# Amp Ref + Driven Right Leg (DRL)

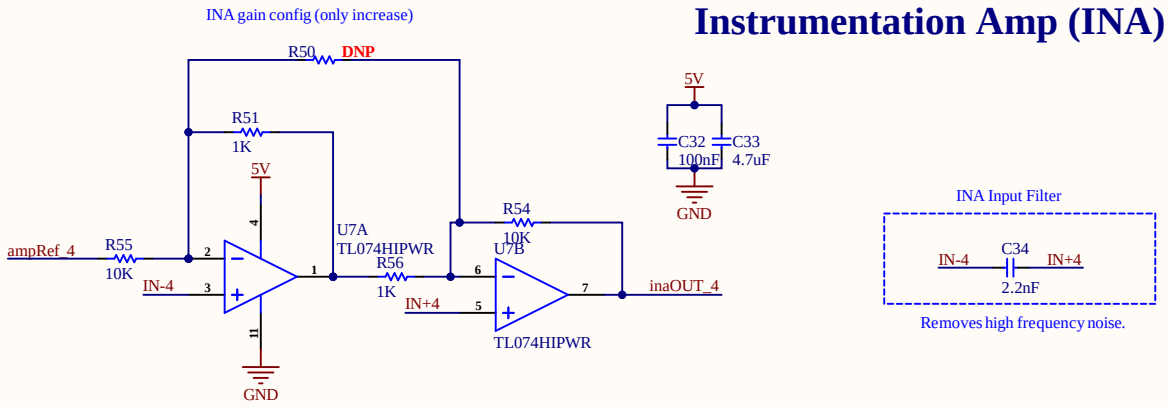



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|---|---|
| Title: Electrode-Sheet3.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/10/2025 | Page Num: 5 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

# Instrumentation Amp (INA)

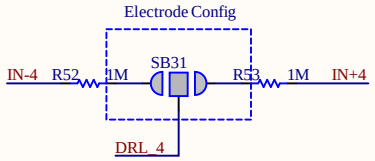






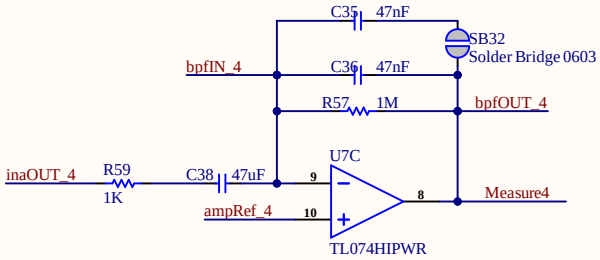
Removes high frequency noise.

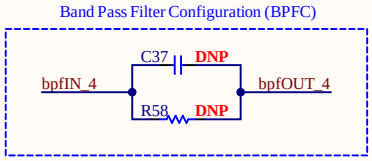
# Electrode Refrence




Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

# 1000x Gain (Bandpass)

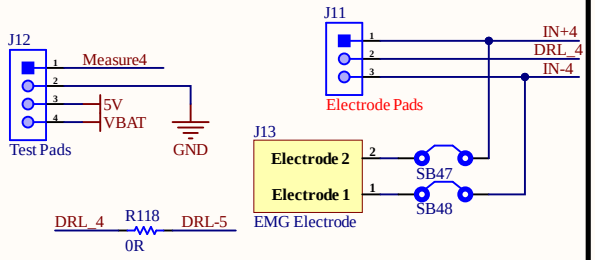


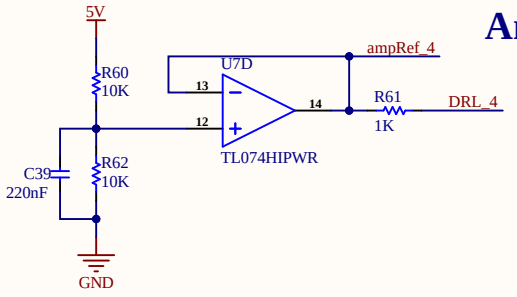

Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

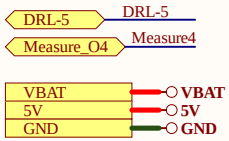
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

# Header Pins / Connectors




# Amp Ref + Driven Right Leg (DRL)

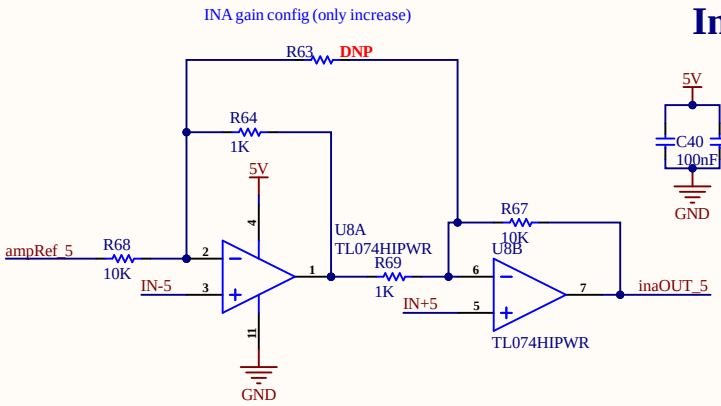


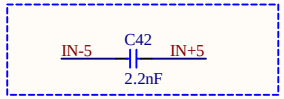




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|---|---|
| Title: Electrode-Sheet4.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/10/2025 | Page Num: 6 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

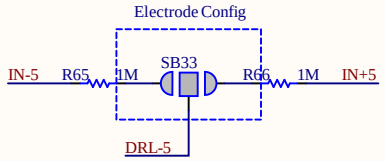
# Instrumentation Amp (INA)



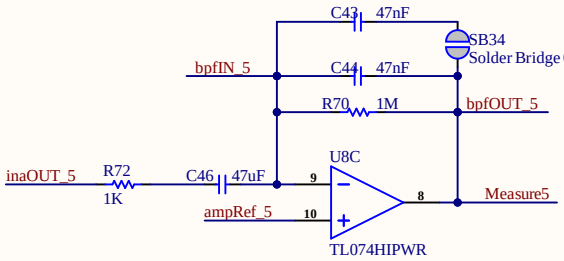

Removes high frequency noise.

# Electrode Refrence

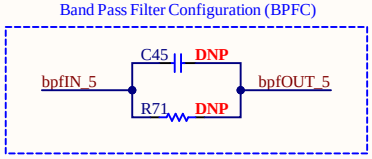



Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

# 1000x Gain (Bandpass)

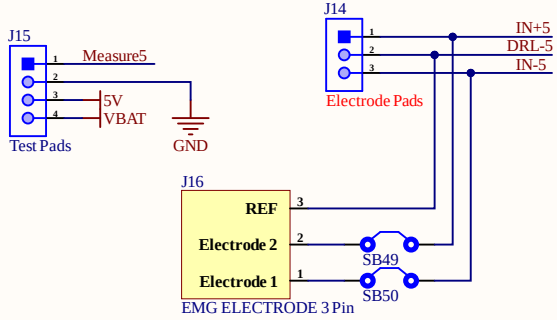



Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

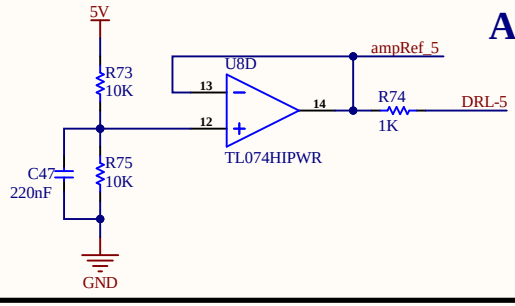


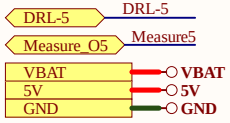

C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

# Header Pins / Connectors




# Amp Ref + Driven Right Leg (DRL)

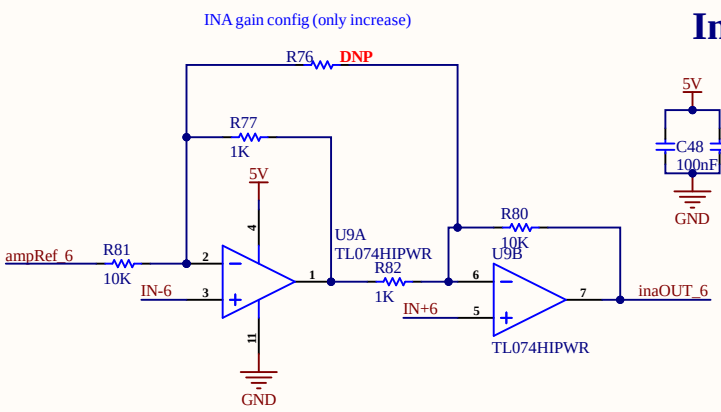


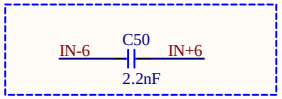




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| | |
|---|---|
| Title: Electrode-Sheet5.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/10/2025 | Page Num: 7 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

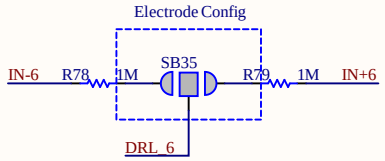
# Instrumentation Amp (INA)



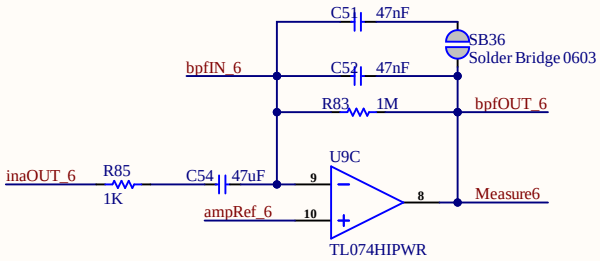

Removes high frequency noise.

# Electrode Refrence

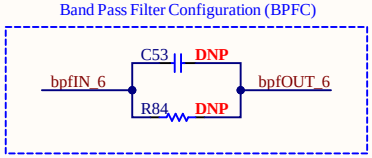



Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

# 1000x Gain (Bandpass)

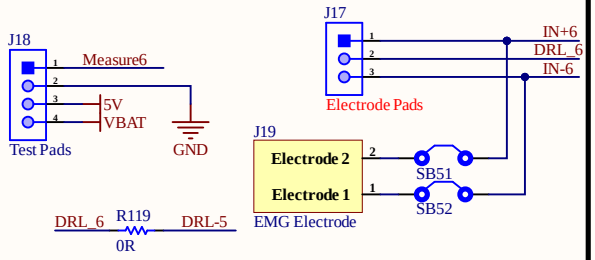



Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

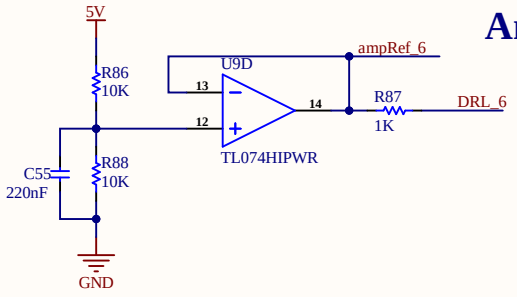


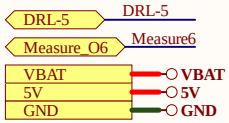

C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

# Header Pins / Connectors




# Amp Ref + Driven Right Leg (DRL)

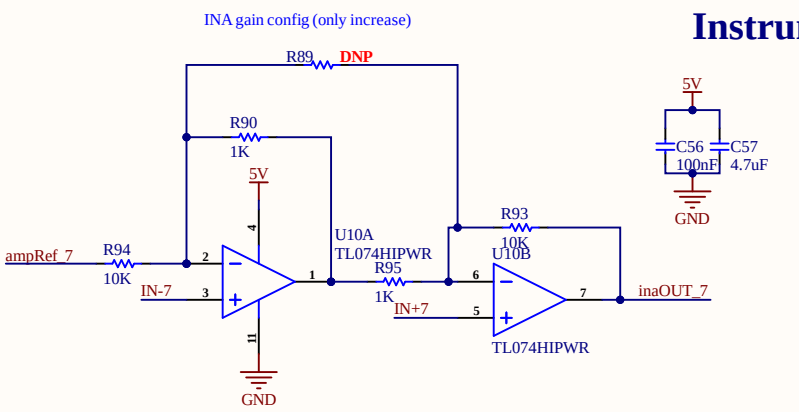


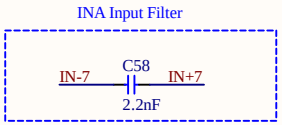




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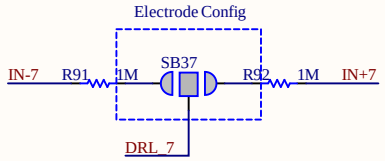
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| Title: Electrode-Sheet6.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/10/2025 | Page Num: 8 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

# Instrumentation Amp (INA)



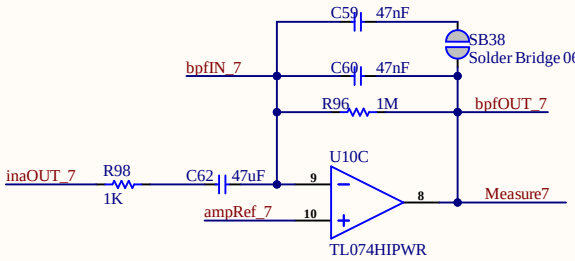



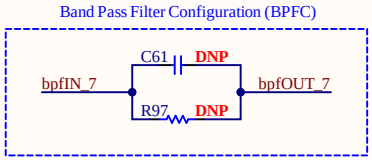

# Electrode Refrence




Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

# 1000x Gain (Bandpass)

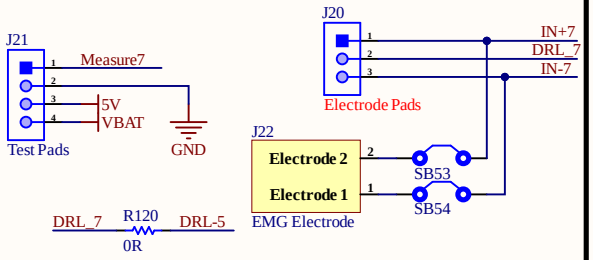


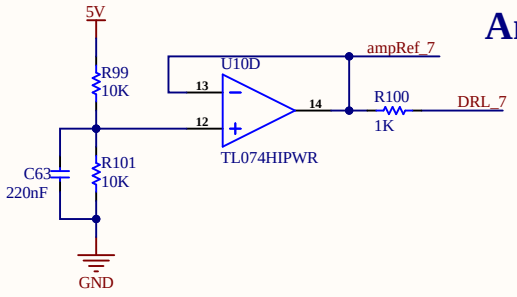

Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

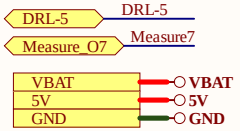
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

# Header Pins / Connectors




# Amp Ref + Driven Right Leg (DRL)

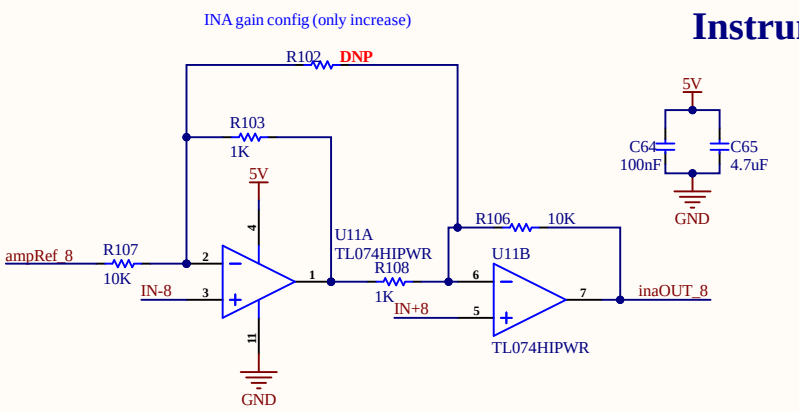


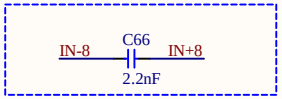




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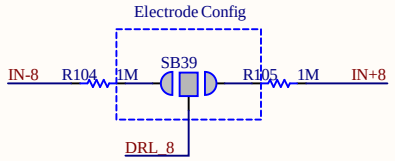
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| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/10/2025 | Page Num: 9 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

# Instrumentation Amp (INA)



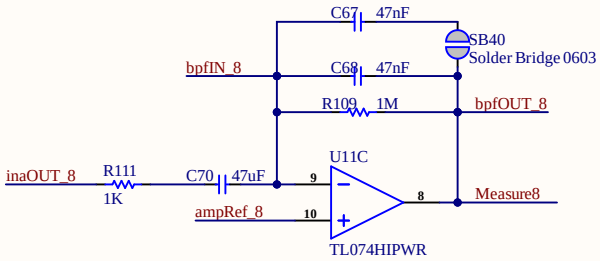



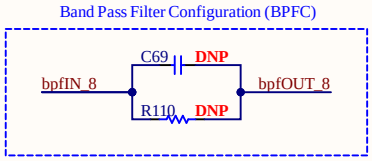

# Electrode Refrence




Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrode will give noisier output than 3 electrode

# 1000x Gain (Bandpass)

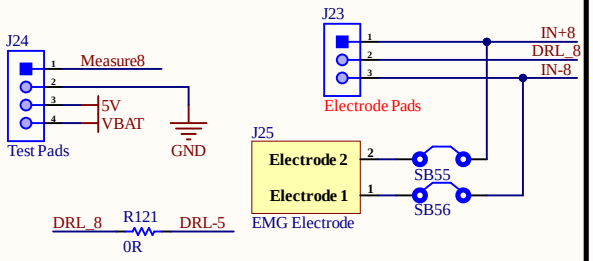


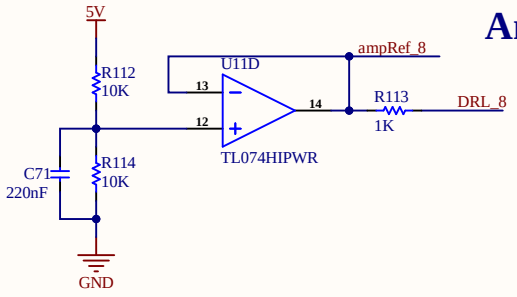

Create solder joint on SB2 to narrow the Band Pass frequency range
Default: Wide input frequency band, use when recording EMG ,EOG
Configured: Narrow input frequency band, use for EEG, EOG, and ECG

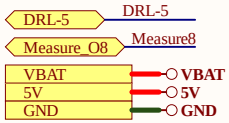
C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

# Header Pins / Connectors




# Amp Ref + Driven Right Leg (DRL)

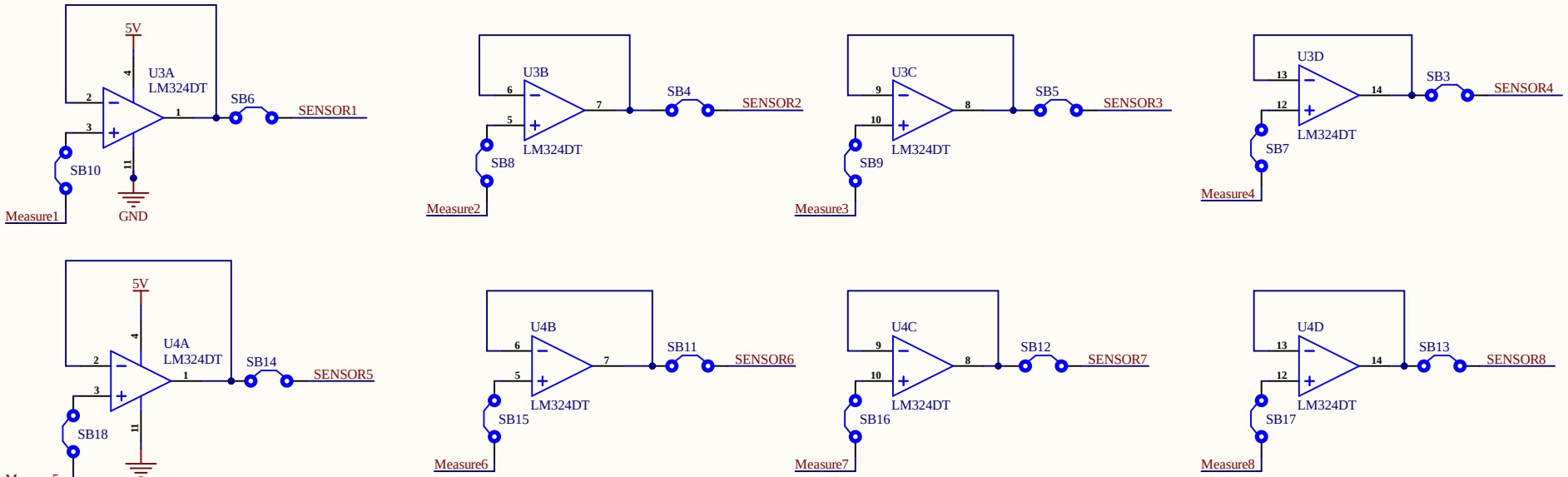


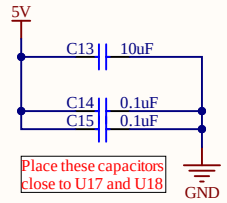

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| Title: Electrode-Sheet8.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/17/2025 | Page Num: 10 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

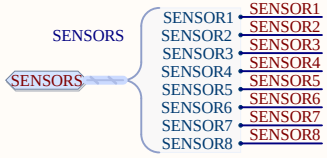
## UNITY GAIN AMPLIFIER FOR IMPEDANCE MATCHING



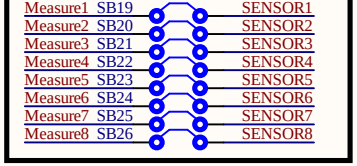

## DECOUPLING CAPS

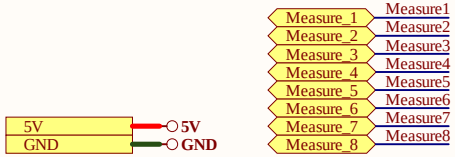



## SESNOR OUT SIGNALS




## SOLDER BRIDGE

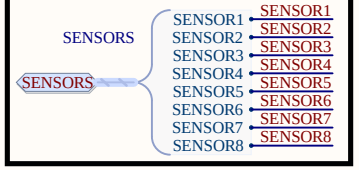


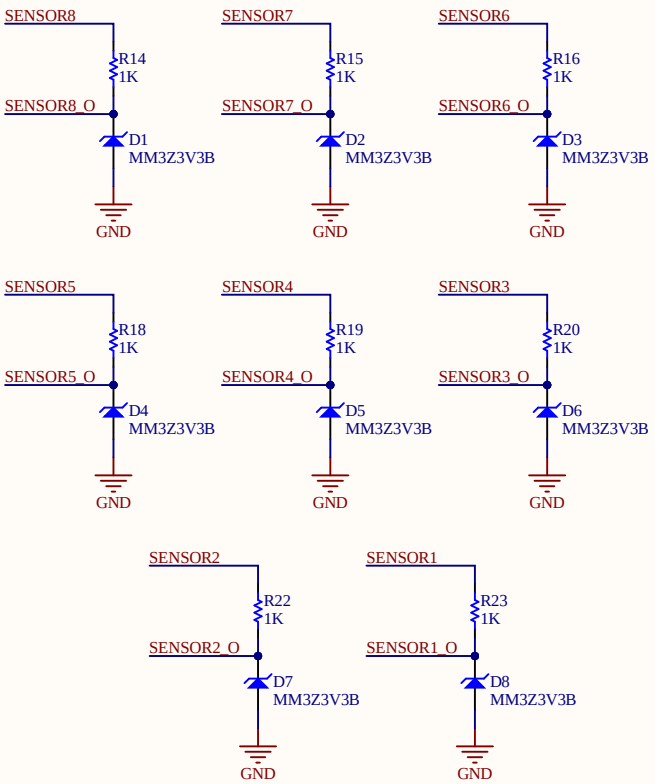

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| | |
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| Title: Opamp-Sheet2.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/10/2025 | Page Num: 11 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

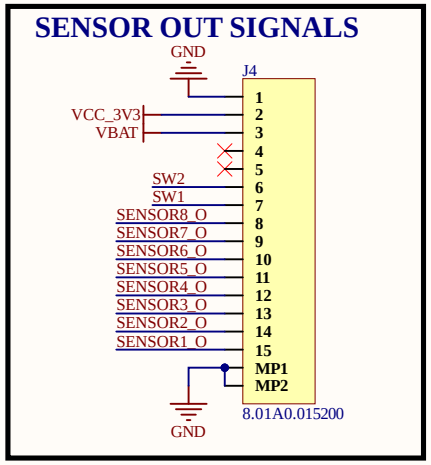
## SESNOR OUT SIGNALS



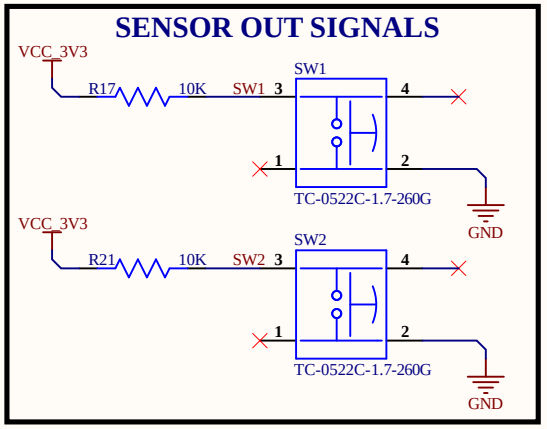

## OVER VOLTAGE PROTECTION

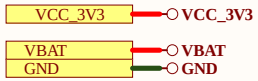



## SENSOR OUT SIGNALS




## SENSOR OUT SIGNALS




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| | |
|---|---|
| Title: Connector-Sheet.SchDoc | |
| File: EMG-Arm_ELC-v1.0.PrjPcb | |
| Date: 2/17/2025 | Page Num: 12 |
| Drawn By: Huzaifa Ali | Page Total: 12 |

| Layer | Name | Material | Thickness | Constant | Flex Layer Stack |
|---|---|---|---|---|---|
| | Top Overlay | | | | |
| | Top Solder | Solder Resist | 0.013mm | 3.5 | |
| 1 | Top Layer | | 0.035mm | | |
| | Dielectric 1 | Flex | 0.020mm | 4.8 | |
| 2 | Bottom Layer | | 0.035mm | | |
| | Bottom Solder | Solder Resist | 0.013mm | 3.5 | |
| | Bottom Overlay | | | | |
| Total board thickness: | | | 0.116mm | | |

